

1 My solutions for exercises

I'm practicing proof writing as well as problem solving here, so the writing style can vary from one solution to another.

1.1 Exercise 2.2.3

1.1.1 (a) $a \geq a$.

Proof. We seek an m such that $a = a + m$. Clearly 0 is such an m , and hence $a \geq a$.

1.1.2 (b) If $a \geq b$ and $b \geq c$, then $a \geq c$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = c + n$. Then we have $a = (c + n) + m$. By the associative law for addition, $a = c + (n + m)$. Therefore $a \geq c$.

1.1.3 (c) If $a \geq b$ and $b \geq a$, then $a = b$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = a + n$. Then we have $a = (a + n) + m$ and $a = a + (n + m)$, and so $n + m$ must be zero. Thus $n = m = 0$ by Corollary 2.2.9. Substituting 0 for m into $a = b + m$, we conclude that $a = b$.

1. Lemma. If $a + m = a$ then $m = 0$.

Proof. $a + m = a = a + 0$. Using the cancellation law gives $m = 0$.

1.1.4 (d) $a \geq b$ iff $a + c \geq b + c$.

Proof.

1. (i) If $a \geq b$ then $a + c \geq b + c$.

Let $a = b + m$ for some natural number m . Then $a + c = (b + c) + m$, hence $a + c \geq b + c$.

2. (ii) If $a + c \geq b + c$ then $a \geq b$.

Since $a + c \geq b + c$, there exists a natural number m such that $a + c = (b + c) + m$. Then $a + c = (b + m) + c$. By the cancellation law $a = b + m$. Hence $a \geq b$.

1.1.5 (e) $a < b$ iff $a++ \leq b$.

Proof. First, suppose $a < b$. Then there exists a natural number m such that

$$b = a + m$$

Since $a < b$ implies $a \neq b$, $m \neq 0$. Thus by Lemma 2.2.10, there is a natural number n such that $n++ = m$. Hence

$$b = a + (n++) = (a + n)++ = (a++) + n$$

Therefore $a++ \leq b$.

Conversely, suppose that $a++ \leq b$. Let m be a natural number such that $b = (a++) + m$. Then

$$b = (a++) + m = (a + m)++ = a + (m++)$$

Suppose, for contradiction, that $a = b$. Then

$$a = a + (m++)$$

and cancelling a gives $m++ = 0$. But that contradicts Axiom 2.3, which says $m++ \neq 0$. Thus $a \neq b$.

Since $b = a + (m++)$, we conclude that $a < b$.

1.1.6 (f) $a < b$ iff $b = a + d$ for some positive number d .

Proof. First, assume that $a < b$. Then $b = a + d$ for some natural number d . Suppose, for contradiction, that $d = 0$. then $b = a + 0 = a$, contradicting the fact that $a \neq b$. Hence $d \neq 0$, and we conclude d is positive.

Conversely, assume that $b = a + d$ for some positive number d . If $a = b$, $d = 0$ by the same argument used in (e), contradicting the fact that $d \neq 0$. Thus $a \neq b$ and we conclude that $a < b$.

1.2 Exercise 2.2.4

Proposition 2.2.13. Let a and b be natural numbers. Then exactly one of the following statements is true: $a < b$, $a = b$, $a > b$.

Proof. We first prove that at most one of the statements is true. If $a = b$, by definition $a < b$ and $a > b$ is false. If $a < b$ or $a > b$, then by definition $a \neq b$. Finally, if $a < b$ and $a > b$, then $a = b$ by Proposition 2.2.12.(c), a contradiction. Hence at most one of the statements is true.

We now prove that at least one of the statements is true. We proceed by induction on a . Consider the base case $a = 0$. Since $b = 0 + b$, we have $0 \leq b$ for all b . Thus either $0 < b$ or $0 = b$, and the base case is done. Now suppose inductively that $a < b \vee a = b \vee a > b$. We need to show that the proposition still holds on $a++$. If $a = b$, we have $a++ = b++ = b + 1$. Since $a++ \neq b$, $a++ > b$. If $a < b$, $a++ \leq b$ by Proposition 2.2.12.(e). Then either $a++ < b$ or $a++ = b$, and in either case we are done. Finally, if $a > b$ then for some natural number m , $a = b + m$. Now $a++ = (b + m)++ = b + (m++)$. Hence $a++ > b$. This closes the induction.

1.3 Exercise 2.2.5.

Proposition 2.2.14 (Strong principle of induction). Let m_0 be a natural number, and let $P(m)$ be a property pertaining to an arbitrary natural number m . Suppose that for each $m \geq m_0$, we have the following implication: If $P(m')$ is true for all natural numbers $m_0 \leq m' < m$, then $P(m)$ is also true. (In particular, this means that $P(m_0)$ is true, since in this case the hypothesis is vacuous.) Then $P(m)$ is true for all natural numbers $m \geq m_0$.

The solution became unnecessarily long here because I tried to give a different proof for each lemma.

1.3.1 Lemma a. $a \neq 0 \rightarrow \exists m(N(m) \wedge a = 1 + m)$.

Proof. Since a is positive, by Lemma 2.2.10 there exists a natural number m such that $a = m++$. Then we have $a = m++ = 0 + m++ = (0 + m)++ = (0++) + m = 1 + m$.

1.3.2 Lemma b. $m \neq 0 \wedge n \neq 0 \rightarrow m + n \neq 1$.

Proof 1. Suppose $m + n = 1$. by Lemma 2.2.10, there exist natural numbers p and q such that $p++ = m$ and $q++ = n$. then $m + n = (p++) + (q++) = ((p + q)++) = 1 = 0++$. by Axiom 2.4. we infer $(p + q)++ = 0$, contradicting axiom 2.3., i.e., $(p + q) \neq 0$. hence $m + n \neq 1$.

Proof 2. Suppose $m + n = 1$. by Lemma 2.2.10, there exists a natural number p such that $p++ = m$. then $m + n = (p++) + n = (p + n)++ = 1 = 0++$. by axiom 2.4 we infer $p + n = 0$. but since n is positive, $p + n \neq 0$. By Proposition 2.2.8, a contradiction. Hence $m + n \neq 1$.

1.3.3 Lemma c. $a < b++ \wedge a \neq b \rightarrow a < b$.

Proof 1 (direct). By Proposition 2.2.12.(f), $b++ = a + n$ for some positive number n . Since n is positive, by Lemma a there exists an m such that $n = m + 1$. Now $b++ = b + 1 = a + n = a + (m + 1) = (a + m) + 1$. Cancelling 1 from both sides yields $b = a + m$. Thus $b \geq a$. Since we also have $a \neq b$, we conclude that $b > a$.

Proof 2 (indirect). Suppose $a < b$ is false. Then by Proposition 2.2.13, $a > b$ must hold. Now choose two positive numbers n and m such that $b++ = a + n$ and $a = b + m$. Now $b++ = b + 1 = a + n = (b + m) + n = b + (m + n)$. Cancelling b from both sides gives $m + n = 1$. But this contradicts Lemma b, since m and n are both positive. Hence $b > a$.

1.3.4 Proof

Proof. Let $Q(n)$ be the property that $P(m)$ is true for all $m_0 \leq m < n$. We prove by induction that $Q(n)$ is true for every natural number n .

If $n = 0$ then $Q(0)$ is vacuously true, since there's no m such that $m < 0$.

Now suppose inductively that $Q(n)$. Let m satisfy $m_0 \leq m < n++$. We must show $P(m)$. We can think of two cases here: $m = n$ and $m \neq n$.

First, consider the case $m = n$. Notice that by the inductive hypothesis, $P(m')$ for all $m_0 \leq m' < n$. It follows that $P(n)$, i.e., $P(m)$, from the hypothesis of the theorem, and we are done.

Now if $m \neq n$, by Lemma c $m < n$. Thus by the inductive hypothesis we infer that $P(m)$.

This closes the induction, and we have proven that $Q(n)$ holds for every natural number n .

Using this result, we now show the conclusion of the theorem, that is, $P(m)$ for all $m \geq m_0$. For every natural number $m \geq m_0$, $Q(m++)$ holds, since $m++$ is a natural number. Together with the fact that $m_0 \leq m < m++$, we conclude that $P(m)$. This demonstrates the strong principle of induction.